1. A Shepherd boy gets bored tending the town's flock. To have some fun he cries out, "wolf" ! even through no wolf is in sight. The villagers run to protect the flock but them get really mad when they realize the boy was playing a joke on them.

One night the shephered boy sees a real wolf approa... the flock and calls out, wolf!" The villagers refuse to be fooled again and stay in their house. [illegible]

TP, FP FN, TN

~~wolf~~ ...

	Y	N
Y	TP	FN
N	FP	TN

- Positive = wolf is actually present
- Nagative = No wolf is present
- Prediction = By cries wolf'

- TP : Real wolf appears + boy cries wolf -> TP
- FP -> No wolf + boy but boy cries "wolf" -> FP
- FN : Real wolf appears + no effective [illegible]

2) Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Indentify TP, FP, FN, TN and calculate the Precision and Recall and F1 score

-

TP	FN
FN	TN

- Total images = 100
- Actual cat images = 30
- Actual non-cat images = 70
- model predicts cat in 25 of 30 cat images
- model predicts no cat in 50 of 70 non-cat images

1) Precision

$$= \frac{TP}{TP+FP}$$

$$= \frac{25}{25+20} \times 100 = \cancel{\frac{25}{45}} \times 100 = 55.56\%$$

2) Recall $= \frac{TP}{TP+FN}$

$$25 \quad \frac{25}{} \times 100 = 83.33\%$$

$$f_1 \text{ Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= 2 \times \frac{0.5556 \times 0.8333}{0.5556 + 0.8333}$$

$f_1 = 66.67\%$

3) A email service Provider wants to classify emails as Spam or not Spam using features:

- Contains "offer" (yes/No)
- Contains "free" (yess/No)
- Email Length (Short/Long)

Smaple Datase

Offer	free	length	class
yes	yes	Short	spam
yes	NO	long	Not Spam
NO	yes	short	spam
NO	NO	Long	Not spam
yes	yes	Long	spam
No	No	Short	Not Spam

1) Calculate Prior Probabilities: P(Spam), P(Not
2) Compute Conditional Probabilities for each feat
3. Using Naive Bayes Classifies of (offer = yes, free
length = Short)

1) P

SP

NO

2)

fo

O

1) Prior Probabilities

spam

$P(spam) = \frac{3}{6} = 0.5$

Not spam

$P(Not\ spam) = \frac{3}{6} = 0.5$

2) Conditional Probabilities

<u>for spam</u>

There are 3 spam emails

offer

· offer = yes → 2

· offer – No – 2

	spam	Not spam
$P(offer = yes/spam) =$	$\frac{2}{3}$	1/3
$P(offer = NO/spam) =$	1/3	2/3
$P(~~offer~~\ free = yes/spam) =$	3/3	0/3
$P(free = No/spam) =$	0/3	3/3
$P(length = ~~No~~\ Short/spam) =$	2/3	1/3
P(...) = 1/ ...		2/3

③ classify the New Email

- offer = yes
- free = No
- Length = short

$P(\text{spam}|x) \propto P(\text{spam}) \times P(\text{yes}|\text{spam}) \times P(\text{spam})$

$= \frac{1}{2} \times \frac{2}{3} \times \frac{1}{3} \times \frac{2}{3}$

$= \frac{2}{27}$

$P(\text{spam}, x) = 0.07407$

Not spam

$P(\text{Not spam}|x) \propto P(\text{Not spam}) \times P(\text{yes}/\text{notspam}) \times P(\text{spam})$

$= \frac{1}{2} \times \frac{1}{3} \times 1 \times \frac{1}{3}$

$= \frac{1}{18}$

$P(\text{Notspam}, x) = 0.05556$

Since

$0.07407 > 0.05556$

(i) weka

Naive Bayes classifier

Attribute	class spam	Not spam
other		
yes	3:0	2.0
NO	2:0	3.0
(Total)	5:0	5.0
free		
yes	4.0	1.0
NO	1.0	4.0
(Total)	5.0	5.0
length		
short	3.0	2.0
length	2.0	3.0
Total	5.0	5.0

~~Graph~~ Model Graph